



GREEN BEAN COFFEE



Two-Stage Sales Analysis

Summary

We propose a comprehensive analysis of April sales data to uncover insights into product performance and wastage. Analysis will uncover purchasing trends by customer demographic, as well as commonly paired items to gain new insights into product recommendations. We will further work to uncover seasonal shopping trends based on publicly available year-long data to better anticipate product demand.

Timeline

Tasks	Due Date	Status
Project Approval	Jun 21, 2025	✓ Resolved ▾
Analysis	Jul 19, 2025	🕒 Ongoing ▾
Dashboards, Powerpoints, and Videos for Management and Analytics Team	Aug 3, 2030	🕒 Ongoing ▾

Objectives

Stage 1:

Store Insights

- *Product Portfolio Analysis and Waste Expenditures*
- *Demographic Insights*
- *Personalized Recommendation Systems*

Stage 2:

Demand Forecasting

- *Anticipate Peak Sales Times*
- *Anticipate Sales Performance by Area*
- *Anticipate Product Performance by Time of Day, Week, Month, and Year*

Desired Outcomes

Objective	Methodology	Benefit/Outcome
Product Portfolio Analysis and Waste Expenditures	Tracking cost per item sold and sales price	Identify key profit drivers
	Track waste percentage per item and lost revenue	Identify opportunities for cost-cutting measures with minimal impact on customer service
Demographic Insights	Track demographic purchases by generation, gender, and neighborhood	Uncover most-profitable customer demographics and untapped market potential
Personalized Recommendation Systems	Build recommendation systems using advanced machine learning techniques	Boost revenue per order by identifying appropriate items to recommend based on other purchases
Anticipate Peak Sales Times	Study publicly available sales data to generalize trends applicable to our stores	Better anticipate sales demands by time of day or week, offering insights into staffing needs and inventory management
	Boost insights with time-series analysis for more accurate predictions	
Anticipate Sales by Area	Mine publicly available data to uncover trends in demand by neighborhood	Offers strategic insight into factors limiting sales and to focus endeavors to boost sales
Anticipate Product Performance by Time of Day, Week, Month, and Year	Use time series analysis to track lead-indicator products that could be used to generalize trends on related items	Build systems that help management better identify when to stock specific items
		Ex: monthly hot chocolate sales could be used to gauge demand for holiday-themed drinks

Resources

- IBM Coffee Shop Data 11.3.1+ (Mock Coffee Company Datasets):

<https://community.ibm.com/community/user/blogs/steven-macko/2019/07/12/beanie-coffee-1113>

- AhmedAbbas75 Github Coffee Shop Sales Data (Mock Year-Round Sales Data):

<https://github.com/Ahmedabbas75/Coffee-Shop-Sales/tree/main>